MIL-E-1/654B
18 April 1968
SUPERSEDING
MIL-E-1/654A
21 January 1959

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 9003

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, UHF, remote cutoff

Outline	---	5-1 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	k, g3, int sd	h	h	p	g2	g3, k, int sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ik	Pg2	Pp	Ehk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	W	W	v	ft
Maximum:	6.9 (see note 1)	275	---	110	---	16	0.33	1.8	90	(see note 3)
Minimum:	5.7	---	-3.0	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	250	-3.0	100	0	---	---	---	---	---

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required

E-50.2 Holding period

E-50.3 Preheating

5. Preparation for delivery

(B) denotes changes

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See note 4	0.65	II	Ic1	0	-1.0	µAdc
1256	Electrode current (anode)		0.65	II	Ib	4.4	8.9	mAdc
1306	Transconductance (1)		0.65	II	Sm	1,300	2,400	µmhos
1306	Transconductance (2)	Ec1 = -35 Vdc	0.65	II	Sm	2	50	µmhos
1231	Emission current	Eb = Ec1 = Ec2 = 15 Vdc (see note 4)	0.65	II	Is	20	---	mAdc
1201	Short and discontinuity detection		0.40	II	---	---	---	---
Appendix D, 20(a), 30	Visual and mechanical inspection criteria		---	---	---	---	---	---
Appendix D, 20(b)	Dimensions	Outline 5-1; dimensions A thru E	1.0	I	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		6.5	L6	If	140	160	mA
1031	Low-frequency vibration	Rp = 10,000 ohms	6.5	L6	Ep	---	400	mVac
(B) 1336	Heater-cathode leakage		6.5	L6	Ihk	---	20	µAdc
1256	Electrode current (screen)		6.5	L6	Ic2	1.4	4.0	mAdc
1311	Anode resistance		6.5	L6	rp	0.45	---	Meg
1306	Transconductance (3)	Ef = 5.5 V	6.5	L6	Sm	1,100	---	µmhos
1211	Insulation of electrodes		4.0	L6	R	10	---	Meg
1331	Direct-interelectrode capacitance	Shield No. 316 Without shield Without shield	6.5	L6	Cgp Cin Cout	--- 2.7 2.1	0.010 4.1 3.9	pf pf pf
1121 and 4.3.2	Base strain		---	---	---	---	---	---
2126	Glass strain		4.0	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501 and 4.7	Life-test provisions (intermittent)	Group A; Ehk = 90 Vdc	---	---	---	---	---	---
4.7.3	Life-test end points (intermittent) (500 hours)	Transconductance (1) Grid emission (see note 2)	--- ---	--- ---	Sm Is	1,200 ---	--- 5.0	µmhos µAdc

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NOTES:

1. Operation of type 9003 in series with other types on supply voltage exceeding 32 volts is not recommended due to the excessive filament surge voltages which will appear on tubes of this type.
2. Ef = 7.5 V; Eb = 250 Vdc; Ec1 = -3 Vdc, Ec2 = 100 Vdc; Rg1 = 0.25 Meg. Read grid emission current with Rg1 = 0 and Ec1 = -45 Vdc while on rack. Duration of test, 48 hours.
3. See 6.5.2.3 for reduced pressure (altitude) rating, and for altitude, maximum peak voltage. (B)
4. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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